

Errata

Corrections to docs/paper.md ("Scaling Laws Do Not Predict Adversarial Robustness in LLM-Based Security Detection Systems"). This is a separate document — the paper itself is not edited. All three items below will be folded into the v2 revision.

Dated 2026-09-05.

1. Meta-Resilience was never computed for Table 1

Section 3.2 declares DABS as five weighted components, including:

Meta-Resilience | 10% | Resistance to adversarial reasoning injection via log fields

Table 1 (§4.1), however, only reports four components — Coverage, Resilience, Hardening, Consistency — with no Meta-Resilience column. The codebase confirms why: no call site that produced the five-model scaling result ever populated `meta_resilience` in a technique result, so the component was always absent for that experiment.

The actual DABS scores in Table 1 were therefore computed on the renormalized 4-component form, with effective weights:

Component	Nominal (paper)	Effective (Table 1, meta absent)
Coverage	30%	33.3%
Resilience	25%	27.8%
Hardening	20%	22.2%
Consistency	15%	16.7%
Meta-Resilience	10%	— (never populated)

This renormalization was never disclosed in the paper. The codebase now names this exact weighting `dabs_v1` (`engine/dabs_scorer.py:WEIGHT_PROFILES`) so it can be reproduced precisely, and every DABS result now records `weight_profile`, `weights_nominal`, and `weights_effective` explicitly instead of silently renormalizing.

Conclusion unaffected. Table 1's numbers do not change — this corrects what the paper says about how they were computed, not the numbers themselves.

2. Two model parameter counts were wrong

Table 1 (§3.3) listed qwen2.5:7b at 7.0B and qwen2.5:14b at 14.0B — rounded, unsourced figures. Verified against the official Hugging Face model cards:

Model	Paper value	Corrected value	Source
qwen2.5:7b	7.0B	7.61B	https://huggingface.co/Qwen/Qwen2.5-7B
qwen2.5:14b	14.0B	14.7B	https://huggingface.co/Qwen/Qwen2.5-14B

The other three model's figures (phi3.5:latest 3.8B, mistral:7b 7.0B, llama3.1:8b 8.0B) were re-verified against their own official cards and are unchanged.

Refitting the same power law ($DABS = a \times P^b$, log-linear regression, same 5 DABS scores from Table 1) with the corrected parameter counts:

Fit	Equation	R²
Published	$DABS = 65.36 \times P^{-0.087}$	0.0557
Corrected	$DABS = 64.67 \times P^{-0.080}$	0.0530

Conclusion unaffected. The corrected R² (0.0530) is within rounding distance of the published value (0.0557) — parameter count explains essentially none of the variance either way.

3. The weekly automated battle badge never reflected a real result

`.github/workflows/weekly-duel.yml` ran every Monday from 2026-04-25 through 2026-08-31 (21 runs) using Groq model IDs (`llama-3.1-70b-versatile` as Attacker, `mixtral-8x7b-32768` as Defender) that were decommissioned on Groq's side. Every Defender call failed; every technique recorded zero rounds; the workflow's own guard only checked "did any technique avoid raising an exception," which a zero-round battle satisfies, so it committed a 0.0% / 0-0 badge to `README.md` every single week without ever raising an error. This automation is entirely separate from the scaling-law experiment — its output was never read by `docs/paper.md` — so it does not affect any number in the paper. It is listed here because it was discovered during this audit and is a factual correction about a claim the

README made ("Last Weekly Battle") that was never true.

Fixed: the badge is paused (README.md), scripts/weekly_battle.py now refuses to run against a known-decommissioned model ID and fails the job (instead of committing) on any technique error or an all-zero result, and raw weekly output is now saved to output/benchmarks/weekly/ (versioned in git) instead of only a gitignored path.

4. The original runs used live threat-intelligence enrichment, not seed-controlled

Live threat-intelligence enrichment (URLhaus/Feodo, added 2026-04-25) was active during the original 5-model runs and is not seed-controlled; the paper's reproducibility claim (seed=42) therefore did not hold. Benchmarks now run with `--threat-intel off` by default; the mode is recorded in every output JSON.

5. The paper's seed=42 claim was false — seeding did not exist yet when Table 1 was produced

The paper states (§3.3) that all runs used a fixed seed of 42. Seeding was added to the agents on 2026-05-10 (commit e30fdb0), two days after the paper draft (eada793, 2026-05-08) and after the Zenodo DOI was minted (d6af684, 2026-05-09). Verified directly: at the commit that introduced the scaling-law feature and produced Table 1 (226aed4, 2026-05-08), neither AttackerAgent nor DefenderAgent had a seed parameter, and no `ollama.chat()` call passed a "seed" key in its options. The Table 1 results were generated without any seed; the reproducibility claim was false for the published data.

This is the leading candidate explanation for why this audit's reproduction (same 5 models, `--seed 42, --threat-intel off`) does not land on the original Table 1 values: the original run was never deterministic in the first place. A dedicated isolation test (mistral:7b, unseeded, 3 independent runs, see docs/scaling_v2_results.md) confirmed unseeded variance is real and large — SD=5.07 across mistral:7b's 3 runs, SD=1.95 across llama3.1:8b's 3 runs — but

not sufficient on its own to fully explain the reproduction gap for every model (§3a of docs/scaling_v2_results.md). This result motivated expanding the grid to n=13 models to get a more reliable scaling-law estimate; see the replication summary below.

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Addendum — Replication results (v1.1 Zenodo update)

Dated 2026-09-07.

The isolation test and full replication queued above are complete; see

docs/scaling_v2_results.md (commit 98d5fe9) for full detail. Summary:

Re-running the original five models under a seeded, network-isolated pipeline did not reproduce Table 1 (deviations of 7-33% per model, model ranking scrambled), consistent with the original single unseeded runs measuring run-to-run noise rather than a stable model property. Expanding the grid to n=13 models (1.2B-117B total parameters, including a domain-specific security model, Foundation-Sec-8B) and refitting under this corrected protocol yields $R^2=0.28$ ($p=0.058$) — a numerically positive but statistically non-significant trend (the n=12 subgrid alone, before adding the security model, reaches $R^2=0.31$, $p=0.050$ — also non-significant, and adding the 13th model moves the fit slightly further from significance, not closer). The original paper's central claim ("scaling laws do not predict adversarial robustness") is treated as not surviving replication: the true result is that a single unseeded run of this size and design cannot support a claim in either direction. This is published as a version update (Zenodo v1.1) rather than a retraction — the reproducible experimental artifact and its capacity for self-correction are more valuable to the research community than the original numerical claim.